

# 10th grade

## Learning evidence T2 L2 Decision analysis supplementary material solutions

**C6 Develops decision-making strategies using probabilities and the maximum opportunity criterion.**

### Problem 1 — Community Bookstore Upgrade

**Problem.** A community bookstore is deciding how to update its storefront.

- Alternatives (profits in thousands of USD): Expand reading space; Launch a mobile book cart.
- States of nature and probabilities: High foot traffic (0,55); Low foot traffic (0,45).

Payoff table (given):

| Alternative  | High traffic (0,55) | Low traffic (0,45) |
|--------------|---------------------|--------------------|
| Expand space | 48                  | 12                 |
| Mobile cart  | 32                  | 20                 |

**Question.** Use the maximum opportunity (minimax regret) criterion to choose the best alternative. Then compare the choice to Maximax, Maximin, and Expected Value.

**Solution.**

**(C6) Maximum opportunity criterion (minimax regret).** Best payoff in each state:

$$\text{High traffic: } \max\{48, 32\} = 48$$

$$\text{Low traffic: } \max\{12, 20\} = 20$$

Regret table (best payoff – payoff):

| Alternative  | High traffic   | Low traffic   | Maximum regret |
|--------------|----------------|---------------|----------------|
| Expand space | $48 - 48 = 0$  | $20 - 12 = 8$ | 8              |
| Mobile cart  | $48 - 32 = 16$ | $20 - 20 = 0$ | 16             |

Minimax regret choice: *Expand space*, since its maximum regret is 8 (smaller than 16).

**Comparison with other criteria.**

- **Maximax:** Choose *Expand space* since  $\max(48, 12) = 48$  is larger than  $\max(32, 20) = 32$ .
- **Maximin:** Choose *Mobile cart* since  $\min(32, 20) = 20$  is larger than  $\min(48, 12) = 12$ .
- **Expected Value:** Favors *Expand space* because the higher payoff in the high-traffic state dominates when weighted by the given probabilities.

## Problem 2 — Cooperative Packing Line Choice

**Problem.** A dairy cooperative is choosing a packing line. The profit payoff table below was obtained from estimated revenues and costs under each order size.

- Alternatives: Automated packing line; Manual packing line.
- States of nature (order size): High order (0,30); Medium order (0,45); Low order (0,25).

Payoff table (profit in thousand USD, given):

| Alternative | High (0,30) | Medium (0,45) | Low (0,25) |
|-------------|-------------|---------------|------------|
| Automated   | 42          | 28            | 8          |
| Manual      | 30          | 24            | 12         |

**Question.** Apply the maximum opportunity criterion and then compare with Maximax, Maximin, and Expected Value.

**Solution.**

**(C6) Maximum opportunity criterion (minimax regret).** Best payoff in each state:

$$\text{High: } \max\{42, 30\} = 42$$

$$\text{Medium: } \max\{28, 24\} = 28$$

$$\text{Low: } \max\{8, 12\} = 12$$

Regret table (best payoff – payoff):

| Alternative | High           | Medium        | Low           | Maximum regret |
|-------------|----------------|---------------|---------------|----------------|
| Automated   | $42 - 42 = 0$  | $28 - 28 = 0$ | $12 - 8 = 4$  | 4              |
| Manual      | $42 - 30 = 12$ | $28 - 24 = 4$ | $12 - 12 = 0$ | 12             |

Minimax regret choice: *Automated packing*, since its maximum regret is 4 (smaller than 12).

**Comparison with other criteria.**

- **Maximax:** Choose *Automated* since its best payoff is 42 (greater than 30).
- **Maximin:** Choose *Manual* since its worst payoff is 12 (greater than 8).
- **Expected Value:** Favors *Automated* because the high and medium outcomes carry more total probability and higher payoffs.

## Problem 3 — Craft Export Shipping Plan

**Problem.** A cooperative that exports handmade crafts must choose a shipping plan. It can ship by air, ship by sea, or delay shipment. Demand and exchange rates combine into four states of nature:

- S1 (high demand + strong currency),  $P = 0,20$

- S2 (high demand + weak currency),  $P = 0,30$
- S3 (low demand + strong currency),  $P = 0,25$
- S4 (low demand + weak currency),  $P = 0,25$

Payoff table (profit in thousand USD, given):

| Alternative | S1 (0,20) | S2 (0,30) | S3 (0,25) | S4 (0,25) |
|-------------|-----------|-----------|-----------|-----------|
| Air         | 60        | 48        | 20        | 10        |
| Sea         | 50        | 44        | 26        | 18        |
| Delay       | 35        | 32        | 28        | 24        |

**Question.** Apply the maximum opportunity criterion and then compare with Maximax, Maximin, and Expected Value.

**Solution.**

**(C6) Maximum opportunity criterion (minimax regret).** Best payoff in each state:

$$S1: \max\{60, 50, 35\} = 60$$

$$S2: \max\{48, 44, 32\} = 48$$

$$S3: \max\{20, 26, 28\} = 28$$

$$S4: \max\{10, 18, 24\} = 24$$

Regret table (best payoff – payoff):

| Alternative | S1             | S2             | S3            | S4             | Maximum regret |
|-------------|----------------|----------------|---------------|----------------|----------------|
| Air         | $60 - 60 = 0$  | $48 - 48 = 0$  | $28 - 20 = 8$ | $24 - 10 = 14$ | 14             |
| Sea         | $60 - 50 = 10$ | $48 - 44 = 4$  | $28 - 26 = 2$ | $24 - 18 = 6$  | 10             |
| Delay       | $60 - 35 = 25$ | $48 - 32 = 16$ | $28 - 28 = 0$ | $24 - 24 = 0$  | 25             |

Minimax regret choice: *Ship by sea*, since its maximum regret is 10 (smaller than 14 and 25).

**Comparison with other criteria.**

- **Maximax:** Choose *Air* since its best payoff is 60 (largest).
- **Maximin:** Choose *Delay* since its worst payoff is 24 (greater than 10 and 18).
- **Expected Value:** Favors *Sea* because it balances payoffs with moderate risk across the four states.

## Problem 4 — Campus Vending Options

**Problem.** A student council is selecting a vending option for campus events. It can choose smart vending kiosks or pop-up snack stalls. Campus traffic can be high ( $P = 0,40$ ), medium ( $P = 0,35$ ), or low ( $P = 0,25$ ). The profit payoff table below was obtained from estimated revenues and costs under each scenario.

Payoff table (profit in thousand USD, given):

| Alternative   | High (0,40) | Medium (0,35) | Low (0,25) |
|---------------|-------------|---------------|------------|
| Kiosks        | 25          | 12            | -1         |
| Pop-up stalls | 23          | 14            | 5          |

**Question.** Apply the maximum opportunity criterion and then compare with Maximax, Maximin, and Expected Value.

**Solution.**

**(C6) Maximum opportunity criterion (minimax regret).** Best payoff in each state:

$$\text{High: } \max\{25, 23\} = 25$$

$$\text{Medium: } \max\{12, 14\} = 14$$

$$\text{Low: } \max\{-1, 5\} = 5$$

Regret table (best payoff – payoff):

| Alternative   | High          | Medium        | Low            | Maximum regret |
|---------------|---------------|---------------|----------------|----------------|
| Kiosks        | $25 - 25 = 0$ | $14 - 12 = 2$ | $5 - (-1) = 6$ | 6              |
| Pop-up stalls | $25 - 23 = 2$ | $14 - 14 = 0$ | $5 - 5 = 0$    | 2              |

Minimax regret choice: *Pop-up stalls*, since its maximum regret is 2 (smaller than 6).

**Comparison with other criteria.**

- **Maximax:** Choose *Kiosks* since its best payoff is 25 (greater than 23).
- **Maximin:** Choose *Pop-up stalls* since its worst payoff is 5 (greater than -1).
- **Expected Value:** Favors *Pop-up stalls* because the medium and low outcomes have substantial probabilities and higher payoffs than kiosks in those states.

**C7 Uses probabilities and expected value to analyze a decision-making problem.**

## P1 Problem 1 — Single Project under Uncertainty

**Problem.** A firm considers launching a project. Profit depends on the state of nature.

|         | $S_1$ | $S_2$ |
|---------|-------|-------|
| Project | 18    | -4    |

Let  $p$  be the probability of state  $S_1$ , and  $1 - p$  the probability of state  $S_2$ . Determine for which values of  $p$  the project is profitable using Expected Value.

**Solution.**

$$EV = 18p + (-4)(1 - p) = 22p - 4$$

Profitability condition:

$$EV > 0 \Rightarrow 22p - 4 > 0$$

## P2 Problem 2 — Technology Upgrade

**Problem.** A company considers upgrading its technology. Profit depends on the state of nature.

|         | $S_1$ | $S_2$ |
|---------|-------|-------|
| Upgrade | 25    | -10   |

Let  $p$  be the probability of state  $S_1$ . Determine for which values of  $p$  the upgrade is profitable.

**Solution.**

$$EV = 25p + (-10)(1 - p) = 35p - 10$$

Profitability condition:

$$EV > 0 \Rightarrow 35p - 10 > 0$$

## P3 Problem 3 — Two Investment Alternatives

**Problem.** Two investment options depend on the state of nature.

|   | $S_1$ | $S_2$ |
|---|-------|-------|
| A | 22    | 6     |
| B | 16    | 10    |

Let  $p$  be the probability of state  $S_1$ . When is option A better than option B using Expected Value.

**Solution.**

$$EV(A) = 22p + 6(1 - p) = 16p + 6$$

$$EV(B) = 16p + 10(1 - p) = 6p + 10$$

Decision condition:

$$EV(A) - EV(B) > 0$$

$$(16p + 6) - (6p + 10) > 0 \Rightarrow 10p - 4 > 0$$

## P4 Problem 4 — Three Production Plans

**Problem.** A firm must choose among three production plans.

|   | $S_1$ | $S_2$ |
|---|-------|-------|
| A | 30    | 0     |
| B | 20    | 8     |
| C | 14    | 12    |

Let  $p$  be the probability of state  $S_1$ . Determine for which values of  $p$  each plan is optimal.

**Solution.**

$$EV(A) = 30p, \quad EV(B) = 20p + 8(1 - p) = 12p + 8, \quad EV(C) = 14p + 12(1 - p) = 2p + 12$$

Pairwise comparison conditions in  $> 0$  form:

$$EV(A) - EV(B) > 0 \Rightarrow 30p - (12p + 8) > 0 \Rightarrow 18p - 8 > 0$$

$$EV(B) - EV(C) > 0 \Rightarrow (12p + 8) - (2p + 12) > 0 \Rightarrow 10p - 4 > 0$$

$$EV(A) - EV(C) > 0 \Rightarrow 30p - (2p + 12) > 0 \Rightarrow 28p - 12 > 0$$

## P5 Problem 5 — Two Alternatives with Three States

**Problem.** Two strategies depend on three states of nature.

|   | $S_1$ | $S_2$ | $S_3$ |
|---|-------|-------|-------|
| A | 20    | 10    | -5    |
| B | 12    | 14    | 0     |

Let  $p_1 = P(S_1)$ ,  $p_2 = P(S_2)$ , and  $1 - p_1 - p_2 = P(S_3)$ . When is strategy A better than strategy B using Expected Value.

**Solution.**

$$EV(A) = 20p_1 + 10p_2 - 5(1 - p_1 - p_2) = 25p_1 + 15p_2 - 5$$

$$EV(B) = 12p_1 + 14p_2$$

Decision condition:

$$EV(A) - EV(B) > 0 \Rightarrow (25p_1 + 15p_2 - 5) - (12p_1 + 14p_2) > 0 \Rightarrow 13p_1 + p_2 - 5 > 0$$

## P6 Problem 6 — Policy Choice under Three States

**Problem.** A government compares two policies under three states of nature.

|   | $S_1$ | $S_2$ | $S_3$ |
|---|-------|-------|-------|
| A | 18    | 8     | -6    |
| B | 10    | 12    | 2     |

Let  $p_1 = P(S_1)$ ,  $p_2 = P(S_2)$ , and  $1 - p_1 - p_2 = P(S_3)$ . When is policy A better than policy B.

**Solution.**

$$EV(A) = 18p_1 + 8p_2 - 6(1 - p_1 - p_2) = 24p_1 + 14p_2 - 6$$

$$EV(B) = 10p_1 + 12p_2 + 2(1 - p_1 - p_2) = 8p_1 + 10p_2 + 2$$

Decision condition:

$$EV(A) - EV(B) > 0 \Rightarrow (24p_1 + 14p_2 - 6) - (8p_1 + 10p_2 + 2) > 0 \Rightarrow 16p_1 + 4p_2 - 8 > 0$$

Equivalently:

$$4p_1 + p_2 - 2 > 0$$

## P7 Problem 7 — New Service Platform Launch

**Problem.** A company considers launching a digital service. Profit depends on two independent factors: customer adoption and system reliability.

|               | Stable | Unstable |
|---------------|--------|----------|
| High adoption | 28     | 6        |
| Low adoption  | -4     | -12      |

Let  $p$  be the probability of high adoption and  $q$  the probability of system stability. Find  $EV(p, q)$  and determine when launching is profitable.

**Solution.**

$$EV(p, q) = 16pq + 18p + 8q - 12$$

Profitability condition:

$$EV > 0 \Rightarrow 16pq + 18p + 8q - 12 > 0$$

## P8 Problem 8 — Regional Store Opening

**Problem.** A retailer evaluates opening a new store. Profit depends on demand and competitor response.

|               | Mild | Aggressive |
|---------------|------|------------|
| Strong demand | 35   | 12         |
| Weak demand   | −5   | −20        |

Let  $p$  be the probability of strong demand and  $q$  the probability of a mild response. Determine when opening the store is profitable.

**Solution.**

$$EV(p, q) = 28pq + 32p + 15q - 20$$

Profitability condition:

$$EV > 0 \Rightarrow 28pq + 32p + 15q - 20 > 0$$

## P9 Problem 9 — Marketing Strategy Choice

**Problem.** A firm must choose between two marketing strategies. Outcomes depend on market interest and campaign execution.

| State of nature            | A   | B  |
|----------------------------|-----|----|
| High interest, Effective   | 30  | 22 |
| High interest, Ineffective | 10  | 14 |
| Low interest, Effective    | −6  | 4  |
| Low interest, Ineffective  | −14 | −8 |

Let  $p$  be the probability of high interest and  $q$  the probability of effective execution. When is strategy A better than strategy B.

**Solution.**

$$EV_A(p, q) = 20pq + 16p + 8q - 14$$

$$EV_B(p, q) = 10pq + 18p + 12q - 8$$

Decision condition:

$$EV_A - EV_B > 0 \Rightarrow (20pq + 16p + 8q - 14) - (10pq + 18p + 12q - 8) > 0 \Rightarrow 10pq - 2p - 4q - 6 > 0$$

## P10 Problem 10 — Investment Portfolio Selection

**Problem.** An investor must choose between two portfolios. Outcomes depend on economic growth and interest rates.

| State of nature           | $X$ | $Y$ |
|---------------------------|-----|-----|
| Strong growth, Low rates  | 26  | 20  |
| Strong growth, High rates | 12  | 16  |
| Weak growth, Low rates    | 2   | 6   |
| Weak growth, High rates   | -10 | -4  |

Let  $p$  be the probability of strong growth and  $q$  the probability of low interest rates. When is portfolio X better than portfolio Y.

**Solution.**

$$EV_X(p, q) = 22pq + 14p + 12q - 10$$

$$EV_Y(p, q) = 14pq + 20p + 10q - 4$$

Decision condition:

$$EV_X - EV_Y > 0 \Rightarrow (22pq + 14p + 12q - 10) - (14pq + 20p + 10q - 4) > 0 \Rightarrow 8pq - 6p + 2q - 6 > 0$$